19. Certified equilibria and conditional persistence in mass-action networks

Pure Thought research portfolio

Revised 17 September 2026

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

A list of chemical reactions can settle into different long-term concentration patterns. Some networks have one resting state; others can behave like a switch, with two stable states separated by an unstable one. This problem asks whether we can find every positive equilibrium of a small, explicitly specified network and prove which are locally stable. It also distinguishes never reaching zero at a finite time from genuinely avoiding extinction in the long run.

Key terms

- **Reaction network** — A set of chemical species and rules converting specified combinations of reactants into products.
- **Mass-action kinetics** — Rate laws in which each reaction rate is proportional to powers of its reactant concentrations, multiplied by a rate constant.
- **Stoichiometry matrix S** — A table describing how each reaction changes the amount of each species.
- **Conservation law** — A combination of species amounts that remains constant under the network's reactions.
- **Compatibility class** — The concentration states accessible under the same conserved quantities and stoichiometric constraints.
- **Equilibrium or steady state** — A concentration pattern where the net rate of change is zero.
- **Local stability** — The tendency of states starting sufficiently near an equilibrium to remain nearby, with asymptotic stability also implying return toward it.
- **Multistationarity and multistability** — Having multiple equilibria, and having multiple stable equilibria, respectively.
- **Persistence** — A long-time condition preventing each initially present species from approaching zero along the trajectory.

Why this matters

Stable switches and coexistence are important organizing ideas in chemistry and biological regulation. An exact small-network analysis can establish whether a proposed mechanism even supports those behaviors. It also makes numerical simulation more interpretable: a trajectory can miss another equilibrium or appear stable simply because it has not been followed long enough.

What progress would mean

Progress would give complete equilibrium information for the stated network, rates and conserved totals. A theorem covering many rate choices would be stronger; identifying a switch in a model would not by itself establish its role in a living system.

Core question

For a finite mass-action network with specified rates and conservation totals, can all positive equilibria in one compatibility class be isolated and their local stability certified?

Scope and assumptions

Input integer reactant/product complexes, positive rational rate constants, and positive rational initial totals. Use small networks first, initially at most four species and eight reactions. Distinguish fixed-parameter root analysis from theorems quantified over all rates.

Keep persistence as a separate theorem-driven module. Catalytic RAF structure belongs primarily to Challenge 29 and does not by itself determine mass-action stability.

Mathematical target

Let S be the stoichiometric matrix, f(x)=Sv(x), v_r=κ_r x^{y_r}, and let rows of L span ker S^T. Solve f(x)=0, Lx=Lx_0, x>0. Remove dependent equations before root isolation. Deficiency is δ=number of complexes−number of linkage classes−rank S.

For an isolated equilibrium, test the Jacobian on im S, not on the full space with conservation-induced zero eigenvalues. Multistationarity means at least two positive equilibria in the same class; multistability additionally needs stability. Persistence means liminf_{t→∞}x_i(t)>0 and requires global hypotheses and boundedness checks.

Required outputs and proof obligations

Return exact network data, deficiency/reversibility facts, conservation laws, isolating boxes with existence/uniqueness proofs, and coverage of the admissible domain. A positive-dimensional equilibrium set is a separate valid outcome. Persistence output must name a sufficient theorem and check every hypothesis, or return unresolved.

Validation and rejection controls

Use A⇌B as a deficiency-zero benchmark. Use the one-species open network ∅⇌X and 2X⇌3X with rates giving f(x)=6−11x+6x^2−x^3; isolate x=1,2,3 and distinguish two stable roots from three equilibria. Include a boundary-approaching trajectory example to distinguish positivity from persistence.

Milestones and research extension

Baseline: exact structural facts and complete small fixed-parameter root enumeration. Strong: certified local stability and one applicable persistence theorem. Research extension: parameter-region proofs or a new structural criterion, with quantifiers stated explicitly.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Angeli, De Leenheer and Sontag, A Petri Net approach to persistence — Checkable sufficient conditions for persistence, with hypotheses.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$\dot{x} = Sv(x), \qquad v_j(x) = k_j \prod_i x_i^{\alpha_{ij}}, \qquad x \in (x_0 + \operatorname{im} S) \cap \mathbb{R}^n_{\geq 0}.$$

$$f(x) = 6 - 11x + 6x^2 - x^3 = -(x-1)(x-2)(x-3), \qquad (f'(1), f'(2), f'(3)) = (-2, 1, -2).$$

The scalar benchmark has two locally stable positive equilibria and one unstable equilibrium. General equilibrium enumeration needs complete coverage within the compatibility class, boundary treatment, and a reduced Jacobian. Positivity invariance, persistence and multistability are different claims.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — persistence shortcut (32, 125–131).** Remaining positive at finite time is not persistence, and the stated unrestricted siphon "if and only if" is not a general algorithmic criterion. A sufficient theorem must carry its kinetic and boundedness assumptions.

2. **Major — multistationarity versus stability (27, 1102).** Multiple equilibria need not all be stable. The bistable Schlögl example typically has three positive equilibria, two stable, rather than the required two equilibria. The new exact polynomial benchmark makes this distinction checkable.
3. **Major — algebraic completeness (41, 55).** A Gröbner basis does not automatically isolate all positive real roots or eliminate continuous solution sets. Conservation classes and positivity restrictions are essential.
4. **Major — parameter independence (57–58).** Network deficiency is structural, but most dynamical conclusions depend on rates and totals unless a theorem quantifies over them. The rewrite separates those claims and avoids promised scalability to hundreds of species.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For a finite mass-action network with specified rates and conservation totals, can all positive equilibria in one compatibility class be isolated and their local stability certified?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: exact structural facts and complete small fixed-parameter root enumeration. Strong: certified local stability and one applicable persistence theorem. Research extension: parameter-region proofs or a new structural criterion, with quantifiers stated explicitly.

Sources supporting the corrections

- Angeli, De Leenheer and Sontag, A Petri Net approach to persistence — Checkable sufficient conditions for persistence, with hypotheses.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What “pure thought” means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means “not excluded”; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.